Equivalence Relation

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Definition: A relation $R \subseteq A \times A$ is called an equivalence relation if

① $(a,a) \in R \quad \forall a \in A$

② $(a,b) \in R \Rightarrow (b,a) \in R$

③ $(a,b), (b,c) \in R \Rightarrow (a,c) \in R$

examples: R is equality $\quad (a,b) \in R \Leftrightarrow a = b$

R is nothing $\quad (a,b) \in R \quad \forall a,b \in A \rightarrow R = A \times A$

$A = \{ \text{all diffable function } f : \mathbb{R} \to \mathbb{R} \}$

$f \sim g \Leftrightarrow f' = g'$

Definition: Given an equivalence relation $R \subseteq A \times A$ the equivalence class of $a \in A$ is

$$[a] := \{ b \in A \mid (a,b) \in R \}$$

examples: $A, = \quad [a] = \{ b \in A \mid a = b \} = \{ a \}$

A, R-nothing $\quad [a] = \{ b \in A \mid (a,b) \in R = A \times A \} = A$

$A = \{ \text{diff functions} \mid fRg \Leftrightarrow f' = g' \}$

$[f] = \{ g \in A \mid f' = g' \} = \{ g \in A \mid (f-g)' = 0 \}$

$= \{ f + c \mid c \in \mathbb{R} \}$